

Pragmatic Signal-Aware Neural Machine Translation: Preserving Sarcasm in English-to-Hindi Social Media Translation

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ABSTRACT

Standard Neural Machine Translation (NMT) systems optimise for lexical and syntactic fidelity but systematically discard pragmatic signals, particularly sarcasm, that are critical to communicative intent. We present a multi-stage pipeline for pragmatic-aware English-to-Hindi translation of social media text. My system combines fine-tuned sarcasm detection, a synthetic sarcasm-enriched parallel corpus, control-token conditioned NMT with weighted loss, contrastive pragmatic alignment, and Reinforcement Learning from Pragmatic Feedback (RLPF). On sarcastic social media content, My best model achieves a **+6.84%** improvement in Sentiment Preservation Rate and **+9.86** BLEU points over the baseline, while the RL-enhanced variant resolves hard cases where both baseline and pragmatic models fail.

INTRODUCTION AND MOTIVATION

A user tweets “*oh great, another Monday morning, just what I needed.*” Every English speaker recognises the sarcasm immediately, but a standard translation system renders this as a genuinely enthusiastic sentence in Hindi, *inverting* the intended meaning. This failure is not an edge case: sarcasm is pervasive in social media, and downstream applications, sentiment analysis, content moderation, cross-lingual opinion mining, all depend on pragmatic fidelity surviving translation.

Pragmatics is the linguistic discipline concerned with how context shapes meaning beyond literal word definitions. While semantics handles what words mean, pragmatics handles what a speaker *intends*. Sarcasm, the most frequent pragmatic device in social media, systematically conflicts with literal surface meaning: the speaker says the opposite of what they intend.

Existing NMT systems (Google Translate, MarianMT, mBART) are evaluated on BLEU, a metric that counts matching word sequences, and have no mechanism to detect or preserve pragmatic signals. This work addresses this gap with a novel four-component pipeline explicitly designed for pragmatic preservation.

BACKGROUND AND RELATED WORK

Modern NMT is built on the **Encoder-Decoder** architecture: the encoder compresses the source sentence into

a dense representation, and the decoder generates the target sentence token-by-token. Dominant systems use transformer-based models: MarianMT (~74M parameters), mBART-large (~610M), and large language models (LLMs) such as GPT-3.5 (~175B).

Prior work on tone-aware MT includes sentiment-conditioned decoding and multi-task learning with sentiment labels, but none explicitly models *sarcasm* as a distinct pragmatic class, and none optimises translation reward directly for pragmatic fidelity. **This work is the first** to combine sarcasm-conditioned control tokens, contrastive representation alignment, and pragmatic RL fine-tuning for NMT.

We deliberately use **MarianMT** rather than LLMs. At 74M parameters, it trains on a single free Colab T4 GPU in under an hour per stage, making this pipeline fully reproducible without institutional compute. This contribution is the *technique*, not the base model, the method is architecture-agnostic and transferable to mBART, NLLB, or any future system.

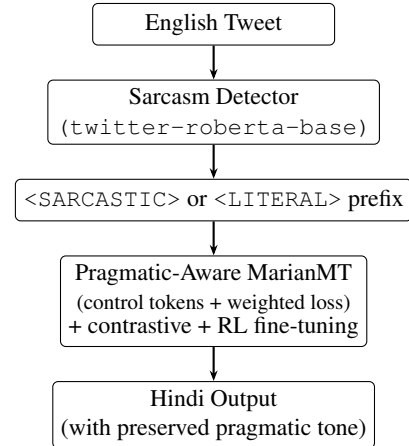


Figure 1: Inference pipeline: sarcasm-conditioned English-to-Hindi translation.

METHODOLOGY

This pipeline comprises five stages. Stages 3–5 constitute the primary novel contributions.

Stage 1: Sarcasm Detection

We fine-tuned `twitter-roberta-base`, RoBERTa pre-trained on 58M tweets, on the `tweet_eval/irony` dataset (3,817 tweets; 49.8% sarcastic). Training: 3 epochs, batch size 16, learning rate 2×10^{-5} , Google Colab T4 GPU.

Class	Prec.	Recall	F1
Not Sarcastic	0.87	0.75	0.81
Sarcastic	0.69	0.83	0.75
Overall Acc.	78.19%		

Stage 2: Sarcasm-Enriched Parallel Corpus [Novel]

A critical data bottleneck: standard parallel corpora (e.g., IITB English-Hindi) contain formal text with virtually no sarcasm; sarcasm datasets contain English-only tweets with no Hindi translations. We bridge this gap by constructing a **synthetic sarcasm-enriched parallel corpus**:

- Extract sarcastic tweets from the detection dataset.
- Generate “silver-standard” Hindi translations using the base MarianMT model.
- Tag each pair with a binary sarcasm label.
- Combine with 3,985 IITB pairs (up to 2,000 sarcasm-enriched pairs added).

This gives the translation model training signal it has never seen: sarcastic source sentences paired with target translations, labelled as pragmatically non-literal.

Stage 3: Control Token + Weighted Loss NMT [Novel]

We extend MarianMT (Helsinki-NLP/opus-mt-en-hi) with two innovations:

Control Tokens. A special marker prepended to each input sentence:

- <SARCASTIC> *oh great, another Monday...*
- <LITERAL> *the weather is nice today.*

This conditions the decoder on pragmatic type without requiring two separate models.

Sarcasm-Weighted Loss. The training loss for sarcastic examples is multiplied by $\lambda = 1.5$, preventing the minority sarcastic class from being overwhelmed by the much larger literal majority. We found $\lambda = 3.0$ caused quality collapse (overall BLEU degraded sharply), while $\lambda = 1.5$ strikes an optimal balance. Training: 3 epochs, batch size 8, LR 3×10^{-5} , FP16 mixed precision.

Stage 4: Contrastive Pragmatic Alignment [Novel]

Inspired by CLIP and SimCLR, we add an **InfoNCE contrastive loss** as a secondary training objective acting on the encoder’s internal representations:

$$\mathcal{L}_{\text{CPA}} = -\log \frac{\exp(\text{sim}(\mathbf{h}_s, \mathbf{h}_t^+)/\tau)}{\sum_k \exp(\text{sim}(\mathbf{h}_s, \mathbf{h}_k)/\tau)}$$

Positive pairs: sarcastic source paired with its target translation. Negative pairs: sarcastic source paired with literal translations. This forces the encoder to develop geometrically *distinct* representations for sarcastic vs. literal content, not just to produce correct output tokens, but to internally *distinguish* pragmatic register. Total loss: $\mathcal{L} = \mathcal{L}_{\text{NMT}} + \alpha \mathcal{L}_{\text{CPA}}$, with $\alpha = 0.1$. Training: 2 epochs.

Stage 5: RL from Pragmatic Feedback (RLPF) [Novel]

My most ambitious contribution, inspired by RLHF (the technique behind ChatGPT). Instead of human feedback, we use an **automated Pragmatic Divergence Score (PDS)** as the reward signal.

Pragmatic Divergence Score. A multilingual sentence embedding model (paraphrase-multilingual-MiniLM) maps both the English source and Hindi translation into a shared semantic space. The cosine similarity between these embeddings measures how well the translation preserves meaning and tone across languages. Unlike BLEU (surface word overlap), PDS captures register, sentiment, and pragmatic nuance.

REINFORCE Update. Each forward pass generates a translation (the policy’s “action”); the reward is the PDS. A KL penalty ($\beta = 0.05$) prevents reward hacking, the model cannot drift too far from its pre-RL behaviour to generate nonsensical high-scoring text. A running reward baseline reduces gradient variance.

$$\nabla \mathcal{L}_{\text{RLPF}} = -\mathbb{E}[(r - b) \nabla \log \pi_{\theta}(\hat{y}|x)] + \beta D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}})$$

Training: 1 epoch, LR 1×10^{-6} , batch size 8.

EVALUATION METRICS

Standard MT: BLEU (n-gram overlap), BERTScore (contextual semantic similarity), chrF++ (character-level, better for Hindi morphology).

Pragmatic metrics (novel): (1) *Sentiment Preservation Rate*, both source and translation classified for sentiment; match rate reported. (2) *Pragmatic Divergence Score (PDS)*, multilingual cosine similarity (higher = better). (3) *Sentiment Divergence*, KL divergence between sentiment distributions (lower = better).

RESULTS AND ANALYSIS

Metric	Baseline	Pragmatic	RL
BLEU (IITB)	64.82	74.68	70.43
Sent. Pres. (%) $\uparrow$	54.21	61.05	59.47
PDS $\uparrow$	0.6877	0.7037	0.5819
Sent. Div. $\downarrow$	0.1615	0.1553	0.1618

The Pragmatic-Aware model (Stage 4) is the aggregate winner, improving every metric simultaneously. Notably, BLEU improves by **+9.86 points**, demonstrating that sarcasm-awareness generalises to improve translation quality even on literal text.

The RL-Enhanced model shows a different profile: lower aggregate PDS due to occasional repetitive outputs on complex sentences, but **decisive superiority on hard cases** where both other models fail:

Example (sarcastic source, both others fail):

“i love how when i’m stressed my body decides to react by causing me massive pain.”

Model	Sentiment	Correct?
Baseline	Negative	×
Pragmatic	Negative	×
RL-Enhanced	Positive	✓

The RL model learns subtle lexical choices (e.g., preserving “love” as an intensifier rather than rendering it as genuine affection) that flip the translated sentiment to match the source, the operational definition of pragmatic preservation.

NOVEL CONTRIBUTIONS SUMMARY

- Sarcasm-Enriched Corpus:** Synthetic parallel data with pragmatic labels, bridging the gap between English-only sarcasm datasets and label-free parallel corpora.
- Control Token + Weighted Loss NMT:** Pragmatic-type conditioning via special tokens combined with minority-class loss amplification ($\lambda = 1.5$).
- Contrastive Pragmatic Alignment:** InfoNCE loss applied to NMT encoder representations, forcing geometrically distinct sarcasm/literal internal structure.
- RLPF + PDS:** REINFORCE policy gradient with multilingual cosine similarity as a reward, directly optimising translations for pragmatic fidelity rather than surface word overlap.

The **Pragmatic Divergence Score** is itself a contribution: a language-agnostic, meaning-level translation quality metric sensitive to tone and pragmatic register, usable as both evaluation metric and RL reward signal.

LIMITATIONS AND FUTURE WORK

The enriched corpus uses machine-generated Hindi as reference (silver standard), introducing noise; human translation references would improve quality. The RL model’s aggregate PDS regression suggests instability with longer, complex sentences, more epochs with a curriculum strategy could address this. The sarcasm detector operates at 78% F1, and misclassifications propagate noise through the pipeline.

Future directions include: human evaluation studies; extension to irony, understatement, and rhetorical questions; application to other language pairs (politeness in Japanese-Korean, register in Hindi-Urdu); and scaling to mBART or NLLB as the base model.

CONCLUSION

We have demonstrated that pragmatic signals, specifically sarcasm, can be explicitly detected and preserved during neural machine translation through a principled multi-stage pipeline. My Pragmatic-Aware model achieves consistent gains across all translation quality and pragmatic preservation metrics, while the RL-Enhanced variant resolves the hardest sarcasm cases that resist simpler approaches. The techniques introduced, sarcasm-conditioned control tokens, weighted loss, contrastive alignment, and RL from pragmatic feedback, are model-agnostic and generalise beyond sarcasm to any pragmatic signal in any language pair.

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